

Homework #11

Solutions

CHE 210, Fall '01
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GTU

① Murphy 6.14

For CO_2 :

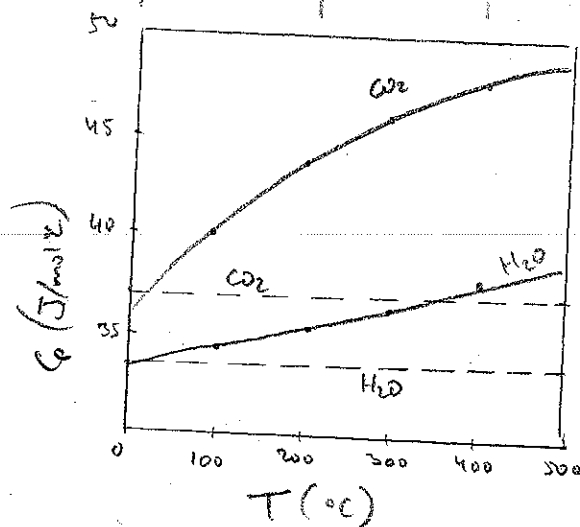
$$C_p = 19.8 + .07344T - 5.602 \times 10^{-5}T^2 + 1.17115 \times 10^{-8}T^3$$

For $\text{H}_2\text{O}(\text{v})$

$$C_p = 32.24 + .001924T + 1.0552 \times 10^{-5}T^2 - 3.596 \times 10^{-9}T^3$$

We can make a C_p vs. T table and then a plot by hand, or use a graphing utility such as excel.

$C_p (\text{J/mol}^\circ\text{C}) \backslash T (^\circ\text{C})$	100	200	300	400	500
$\text{CO}_2(\text{g})$	40.01	43.24	45.69	47.42	48.50
$\text{H}_2\text{O}(\text{v})$	34.24	35.13	36.13	37.22	38.37



② As is evident from the plot, heat capacities of both substances increase significantly with both substances. This is demonstrated by the solid lines representing the more accurate mathematical expressions for the C_p 's. The dashed lines represent the approximate, room-temp values, but as we can see, both become increasingly less accurate as we deviate from room temp.

The rule of thumb on p. 523 tells us that the C_p of a gas is about $15 \text{ J/g} \cdot ^\circ\text{C}$. To use this estimate, we first convert it to moles

$$\text{CO}_2: \frac{15}{\text{g} \cdot ^\circ\text{C}} \cdot \frac{44 \text{ g}}{\text{mol}} = \underline{44 \text{ J/mol} \cdot ^\circ\text{C}}$$

$$\text{H}_2\text{O}(v): 1.8 = \underline{18 \text{ J/mol} \cdot ^\circ\text{C}}$$

It then becomes clear that the rule of thumb is a reasonable estimate for CO_2 , but that water vapor obeys poorly, even in its vapor phase... (why?) ☺

(2) Murphy 6.20 (parts a, b, c, d, e, h, & i)

a) System: turbine cont.

$$\Delta PE = 0 \text{ (no } \Delta h)$$

$$\Delta KE = 0 \text{ (no } \Delta v)$$

$$\dot{W}_s \neq 0$$

$$\dot{Q} \neq 0$$

$$\dot{m}_0 \vec{H}_0 - \dot{m}_1 \vec{H}_1 = \dot{Q} + \dot{W}$$

$$\Rightarrow \dot{Q} < 0 \text{ \& } \dot{W}_s < 0$$

exp.: Work is done by the turbine contents on the surroundings & heat is lost...

note: by convention, positive $\dot{W}$ & $\dot{Q}$ terms mean that the energy is leaving the system, i.e. the system is losing energy.

(3)

b) System: fluid stream between inlet & outlet

$$\Delta h = \Delta v = 0$$

$$\Delta PE = \Delta KE = 0$$

$$\dot{W}_s = 0; \dot{Q} \neq 0$$

$$\dot{m}_o \hat{H}_o - \dot{m}_i \hat{H}_i = \dot{Q}$$

$$\Rightarrow \dot{Q} > 0$$

exp.: No work is done by the fluid, but heat is added to the system...

c) System: H_2O in the gate & the turbine

also, $\dot{Q} = 0$
but $\dot{W}_s \neq 0$

- no phase change

$$\Delta P, \Delta T = 0$$

$$\Delta v = 0$$

$$\Delta H = 0$$

$$\Delta KE = 0$$

$$\dot{m}_o \hat{PE}_o - \dot{m}_i \hat{PE}_i = \dot{W}_s$$

$$\Rightarrow \dot{W}_s < 0$$

exp.: the H_2O does work on the surroundings by turning the turbine, thereby losing energy.

And, by convention, energy lost by the system via work gives a negative $\dot{W}_s$ term.

However, $\Delta h \neq 0 \Rightarrow \Delta PE \neq 0$

d) System: oil inside pipeline between pts. A & B

$$\Delta h \neq 0 \Rightarrow \Delta PE \neq 0$$

$$\Delta v = 0 \Rightarrow \Delta KE = 0$$

$$\dot{W}_s \neq 0 \text{ (pump)}$$

$$\dot{Q} \neq 0 \text{ (heat lost due to poor insulation)}$$

$$\dot{m}_o \hat{PE}_o - \dot{m}_i \hat{PE}_i = \dot{Q} + \dot{W}_s$$

$$\Rightarrow \dot{Q} < 0 \text{ \& } \dot{W}_s > 0$$

exp.: Pump adds energy to the system via shaft work $\Rightarrow +\dot{W}_s$; Heat is lost to surrounding due to imperfect pipe insulation $\Rightarrow -\dot{Q}$

Assume that $\Delta T \approx 0$, since heat from friction is lost to surroundings.

4)

e) System: reactor & contents
 Assume - (well-insulated) adiabatic ($\dot{Q} = 0$)
 Little else is given
 Since this is a chemical rxn, we can generally ignore any minor PE & KE changes (rxn related)

Energy Bal:

$$\dot{n}_{N_2,i} \cdot \hat{H}_{N_2,i} + \dot{n}_{H_2,i} \cdot \hat{H}_{H_2,i} = \dot{n}_{NH_3,o} \cdot \hat{H}_{NH_3,o} \Rightarrow \dot{Q} = 0$$

h) Similar to (e), but with heat input from the furnace (surroundings):

$$\dot{n}_o \hat{H}_o - \dot{n}_i \hat{H}_i = \dot{Q} \quad \left(w/ \dot{Q} > 0 \text{ \& } \dot{w}_j = 0 \right)$$

i) System: cup contents:

Assume that the cup is very well isolated; i.e. the system is adiabatic...

Then, the Energy Balance is:

$$\frac{dU_{sys}}{dt} = \dot{m}_{milk,in} \cdot \hat{H}_{milk,in}$$

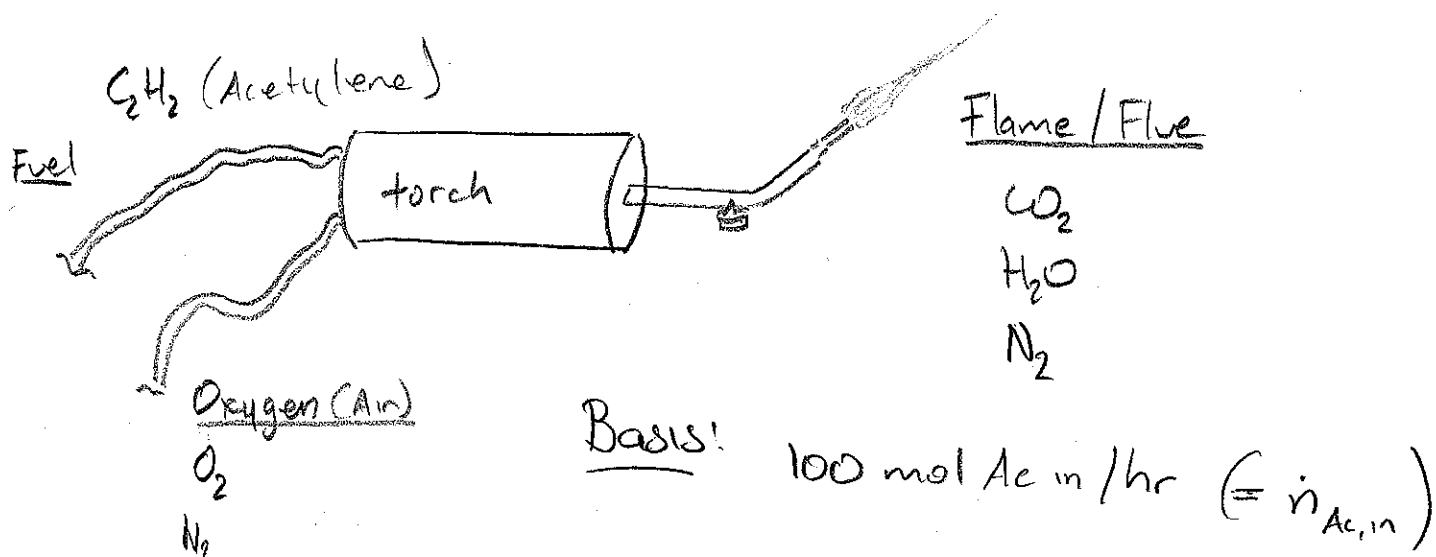
note: in part (i), since the cup is very well isolated, $\dot{Q} = 0$.

However, in real life, this is pretty much impossible, so a small, negative $\dot{Q}$ term would be included...

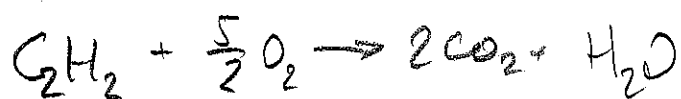
(3) Murphy 6.47

5

Process Flow Diagram:



Balanced Chemical Reaction:



From the feed ratio spec. (just enough O₂ for complete combustion of C₂H₂), we have:

$$\dot{n}_{\text{O}_2,\text{in}} = \frac{5}{2} \dot{n}_{\text{Ac},\text{in}} = \underline{250 \text{ mol O}_2 / \text{hr}};$$

Next, since air is commonly taken to be 21 mol% O₂ & 79% N₂,

$$\dot{n}_{\text{N}_2,\text{in}} = \frac{79}{21} \cdot \dot{n}_{\text{O}_2,\text{in}} = 940.5 \text{ mol/hr} = \dot{n}_{\text{N}_2,\text{out}} \quad (\text{since N}_2 \text{ is an inert gas})$$

⑥

Now, based on the 100% conversion of C_2H_2 & the rxn Stoichiometry, we can write:

$$\dot{n}_{C_2H_2, out} = 200 \text{ mol/hr} \quad \& \quad \dot{n}_{H_2O, out} = 100 \text{ mol/hr}$$

Figuring out the Flame Temperature:

Relative Enthalpies:

1. Take 298 K & 1 atm as reference pt. conditions...
2. Assume that the inlet streams (C_2H_2 & Air) are at or close to room temp ($\sim 298 \text{ K} / 25^\circ \text{C}$)...
3. Obtain necessary Enthalpy / Internal En. data from Table B.3 (Murphy), or other reliable source...

$$\hat{H}_{C_2H_2, in} = -\Delta \hat{H}_{C_2H_2}^{\circ} \text{ (heat of comb. of } C_2H_2) = 1257 \text{ kJ/mol}$$

$$\hat{H}_{O_2, in} = 0 \text{ (reference)}$$

$$\hat{H}_{N_2, in} = 0 \text{ (reference)}$$

4. We can get the outlet "flame" temperature by using C_p 's. Since this is only a rough comparison, we do not need to go through all the trouble of using the polynomial expressions for $C_p(T)$...

We can also neglect any "Enthalpies of Mixins".

(7)

Thus, we have:

$$\hat{H}_{\text{CO}_2, \text{out}} \approx C_{p(\text{CO}_2)} (T_f - T_{\text{ref}}) = \frac{37(T_f - 298)}{1000} = .037T_f - 11 \text{ kJ/mol}$$

$$\hat{H}_{\text{H}_2\text{O}, \text{out}} \approx C_{p(\text{H}_2\text{O})} (T_f - T_{\text{ref}}) = \frac{33.6(T_f - 298)}{1000} = .0336T_f - 10 \text{ kJ/mol}$$

$$\hat{H}_{\text{N}_2, \text{out}} \approx C_{p(\text{N}_2)} (T_f - T_{\text{ref}}) = \frac{29.1(T_f - 298)}{1000} = .029T_f - 8.7 \text{ kJ/mol}$$

5. Consider the flame adiabatic... that way you do not need to consider any loss of heat to the surroundings (i.e. $\dot{Q} \approx 0$); In addition, say that the entire system (flame included) is @ steady-state...

6. Energy Balances:

Acetylene:

$\Delta H_{\text{C}_2\text{H}_2}^\circ$ but $\Delta H_{\text{H}_2\text{O}}^\circ$ subtracted

$$\sum_{\text{in}} \dot{n}_i \hat{H}_i = \sum_{\text{out}} \dot{n}_j \hat{H}_j \Rightarrow 100(12.16) + 250(0) + 940.5(0)$$

$$= 200(.037T_f - 11) + 100(.0336T_f - 10)$$

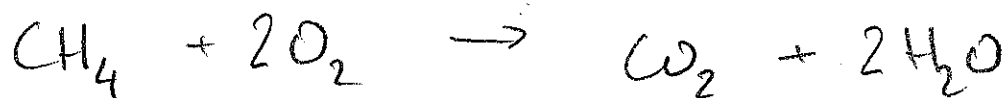
$$\text{Solving for } T_f, \text{ we get: } + 940.5(.029T_f - 8.7)$$

$$T_f(\text{C}_2\text{H}_2) \approx 3496 \text{ K}$$

⑧

We can repeat the same procedure for the methane process (CH_4):

Rxn:



Basis: 100 mol/hr CH_4

$$\dot{n}_{\text{O}_2, \text{in}} = 200 \text{ mol/hr} \quad \dot{n}_{\text{N}_2, \text{in}} = 200 \frac{79}{21} = 752.4 \text{ mol/hr} = \dot{n}_{\text{N}_2, \text{out}}$$

$$\dot{n}_{\text{CO}_2, \text{out}} = 100 \text{ mol/hr} \quad \dot{n}_{\text{H}_2\text{O}, \text{out}} = 200 \text{ mol/hr}$$

Enthalpies: (rel.: 25°C, 1 atm)

$$\hat{H}_{\text{CH}_4, \text{in}} = -\Delta \hat{H}_{\text{c, CH}_4}^\circ = 802.6 \text{ kJ/mol}$$

$$\hat{H}_{\text{O}_2, \text{in}} = 0; \quad \hat{H}_{\text{N}_2, \text{in}} = 0$$

As before,

$$\hat{H}_{\text{CO}_2, \text{out}} \approx .037 T_f - 11 \text{ kJ/mol}; \quad \hat{H}_{\text{H}_2\text{O}, \text{out}} \approx .0336 T_f - 10 \text{ kJ/mol}; \quad \hat{H}_{\text{N}_2, \text{out}} \approx .0298 T_f - 8.745 \text{ kJ/mol}$$

$$\sum \dot{n}_i \hat{H}_i - \sum \dot{n}_j \hat{H}_j \Rightarrow \overset{\Delta \hat{H}_{\text{c, CH}_4}^\circ - \Delta \hat{H}_{\text{f, CH}_4}^\circ}{100(761.6)} = 100(.037 T_f - 11) + 200(.0336 T_f - 10) + 752.4(.0298 T_f - 8.745)$$

Solving for T_f , we get:

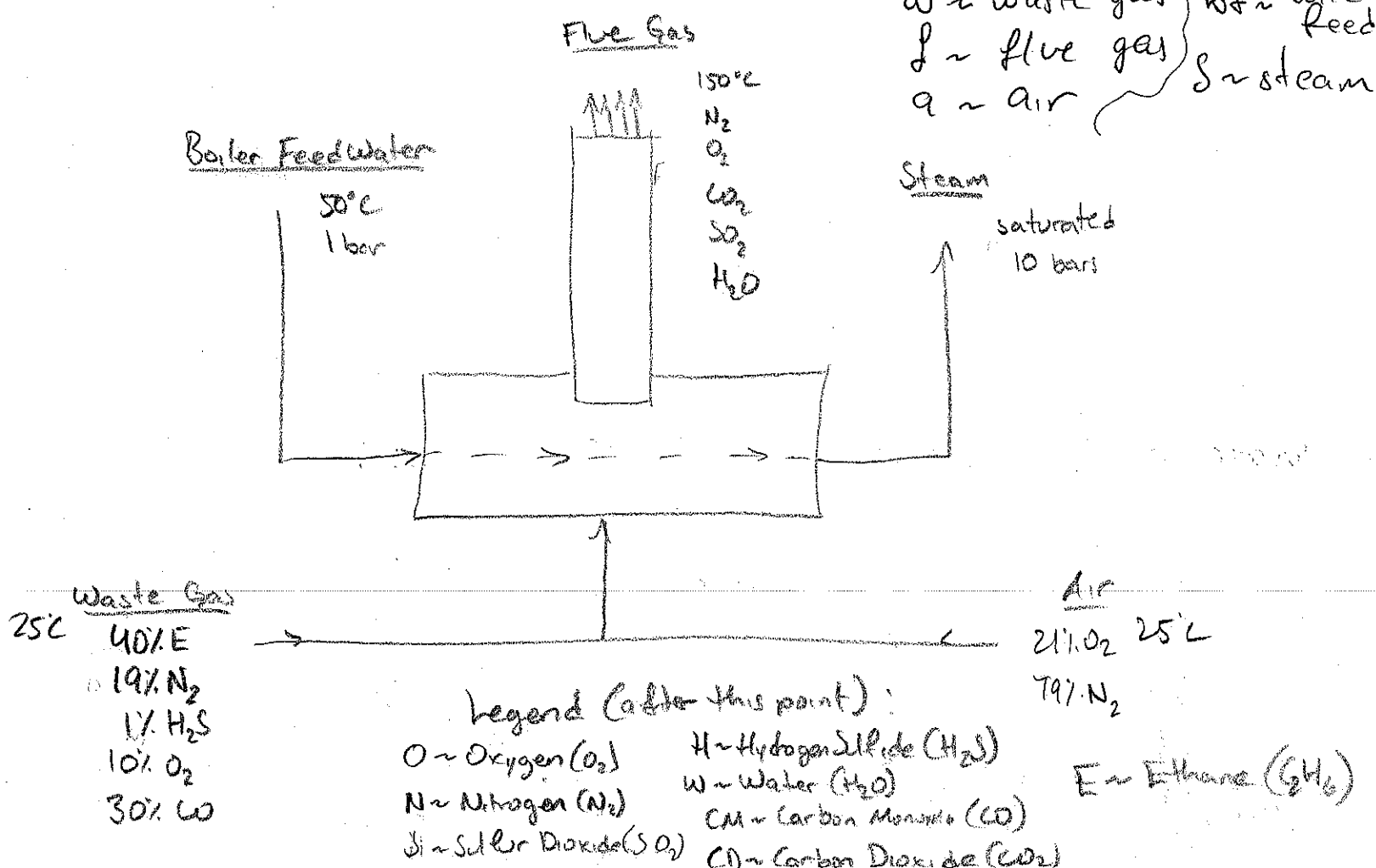
$$T_f \approx 2662 \text{ K}$$

Although the "adiabatic flame approx." is a poor one, we still get a good idea of how much cooler a ^(CH₄) Methane flame will be compared to an Acetylene ^(C₂H₂) flame: $3496 - 2662 = 834\text{K}$ cooler!

(4) Murphy 6.48

Process Flow Diagram

Stream Name legend:
w ~ waste gas } bf ~ boiler feed
l ~ flue gas } s ~ steam
a ~ air



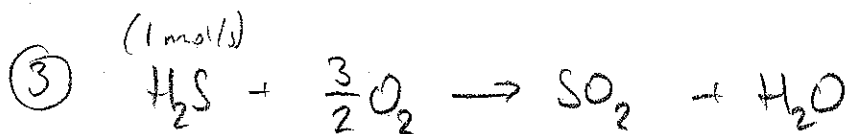
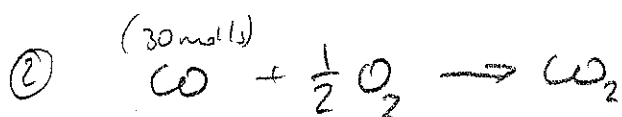
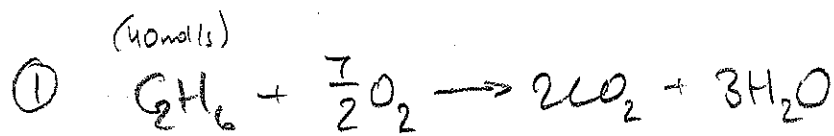
⑩

Base: $\dot{n}_{\text{waste}} = 100 \text{ mol/s}$

This gives:

$$\left. \begin{array}{l} \dot{n}_{E,w} = 40 \text{ mol/s} \\ \dot{n}_{N,w} = 19 \text{ mol/s} \end{array} \right\} \left. \begin{array}{l} \dot{n}_{O,w} = 10 \text{ mol/s} \\ \dot{n}_{H,w} = 1 \text{ mol/s} \end{array} \right\} \dot{n}_{CM,w} = 30 \text{ mol/s}$$

Chemical Reactions:



Oxygen required for complete combustion:

$$\left(\frac{7}{2} \right)^{\textcircled{1}} 40 + \left(\frac{1}{2} \right)^{\textcircled{2}} 30 + \left(\frac{3}{2} \right)^{\textcircled{3}} 1 - 10 \quad \begin{array}{l} \text{included in} \\ \text{waste stream} \end{array}$$

$$= 146.5 \text{ mol/s}$$

Air is in 20% excess:

$O_2: 1.2(146.5) = 175.8 \text{ mol/s}$

$N_2: 175.8 \cdot \frac{79}{21} = 661.3 \text{ mol/s}$

Air Stream

Material Balances (on E, CM, & H):

E: $\dot{n}_{E,f} = 0 = 40 - \dot{f}_1 \Rightarrow \dot{f}_1 = 40 \text{ mol/s}$

CM: $\dot{n}_{CM,f} = 0 = 30 - \dot{f}_2 \Rightarrow \dot{f}_2 = 30 \text{ mol/s}$

H: $\dot{n}_{H,f} = 0 = 1 - \dot{f}_3 \Rightarrow \dot{f}_3 = 1 \text{ mol/s}$

(11)

More Material Balances (based on $\dot{F}_1, \dot{F}_2 \& \dot{F}_3$):
... in exiting flue stream...

$$\dot{n}_{CO, f} = 2\dot{F}_1 + \dot{F}_2 = 80 + 30 = 110 \text{ mol/s}$$

$$\dot{n}_{S, f} = \dot{F}_3 = 1 \text{ mol/s}$$

$$\dot{n}_{W, f} = 3\dot{F}_1 + \dot{F}_3 = 120 + 1 = 121 \text{ mol/s}$$

$$\dot{n}_{N, f} = \dot{n}_{N, w} + \dot{n}_{N, a} = 19 + 661.3 = 680.3 \text{ mol/s}$$

$$\begin{aligned}\dot{n}_{O, f} &= \dot{n}_{O, w} + \dot{n}_{O, a} - \frac{7}{2}\dot{F}_1 - \frac{1}{2}\dot{F}_2 - \frac{3}{2}\dot{F}_3 \\ &= 10 + 175.8 - 140 - 15 - 1.5 \\ &= 29.3 \text{ mol/s}\end{aligned}$$

On to Balancing the Energy:

1. start w/ waste gas & air @ 298 K & 1 atm
2. calculate ΔH_c° (enthalpy change due to combustion @ standard conditions)
3. calculate enthalpy change due to raising the temperature from 298 K to specified flue outlet temperature of $150^\circ\text{C} \equiv 423\text{K}$

note: To finish solving this problem we will follow the step-by-step procedure described above. However, there

⑫ note (cont.):

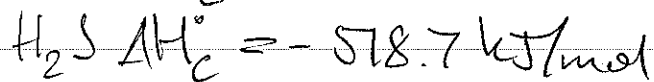
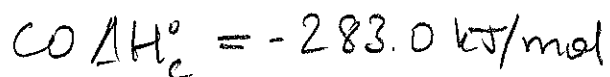
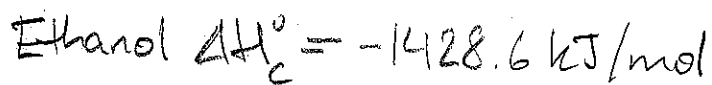
... is definitely more than one way to balance energy, since enthalpy (H) is a "state variable", which means that the path that the process takes (or more precisely, the path that we take while following the process) does not matter. What matters are the starting & the finishing points.

Another, equally valid step-by-step procedure to balance the energy is as follows:

1. start w/ elements @ 298 K & 1 atm
2. calculate enthalpies of the compounds from standard enthalpies of formation
3. correct the enthalpies of the exiting flue gas components for the temperature increase....

Following the first procedure:

list of the required data



$$\text{CO}_2 \quad C_p = 19.80 + .07344T - 5.602 \times 10^{-5} T^2 + 1.7115 \times 10^{-8} T^3$$

$$\text{H}_2\text{O} \quad C_p = 32.24 + .001924T + 1.055 \times 10^{-5} T^2 - 3.589 \times 10^{-9} T^3$$

$$\text{O}_2 \quad C_p = 28.11 - 3.684 \times 10^{-6} T + 1.746 \times 10^{-5} T^2 - 1.065 \times 10^{-8} T^3$$

$$\text{N}_2 \quad C_p = 31.15 - .01357T + 2.680 \times 10^{-5} T^2 - 1.168 \times 10^{-8} T^3$$

$$\text{SO}_2 \quad C_p = 23.85 + .06699T - 4.961 \times 10^{-5} T^2 + 1.328 \times 10^{-8} T^3$$

We can calculate the net enthalpy of the exiting flue stream (kJ/s) as follows:

$$\begin{aligned} H_{f, \text{net}} &= \sum_j \dot{n}_j \Delta \hat{H}_{c,j} + \sum_i \dot{n}_{i, \text{flue}} \int_{T_{\text{in}}}^{T_{\text{out}}} C_{p,i} dT \\ &= (40(-1428.6)) + (30(-283.0)) + (1(-518.7)) \\ &\quad + (110(4.97)) + (121(4.72)) + (29.3(3.73)) + (680.3(3.65)) + (1(5.27)) \\ &= \underline{-62,400 \text{ kJ/s}} = H_{f, \text{net}} \end{aligned}$$

We assume that the system is @ steady-state & realize that the heat transferred to the water in the pipes where the saturated steam (10 bars) is generated is the difference between the input and output net enthalpies:

$$\dot{H}_{\text{out}} - \dot{H}_{\text{in}} = \dot{Q}$$

(14)

Since our reference was the state of the component going in, we have

$$\dot{H}_{out} - \dot{H}_{in} = -62,400 - 0 = -62,400 \text{ kJ/s} = \dot{Q}$$

This flow of heat is what generates the steam.

∴ for the H_2O in the pipes,

$$\dot{H}_{out} - \dot{H}_{in} = +62,400 \text{ kJ/s}$$

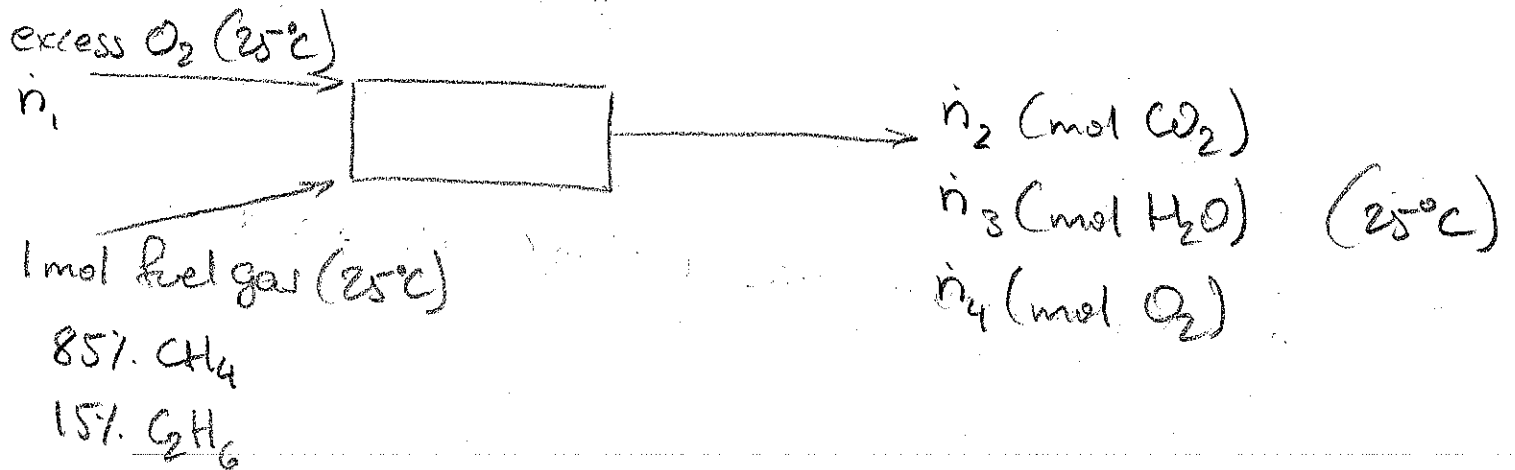
To make the calculations more convenient, we can choose a different reference point for the water. We can take the triple pt. of H_2O as the reference so that we can use the steam tables in the back of the book:

$$\begin{aligned} \dot{m}_{\text{steam}} (\overset{\text{(kJ/kg)}}{\dot{H}_{\text{steam}}} - \dot{H}_{\text{inflowing H}_2\text{O}}) &= \dot{m}_{\text{steam}} (2777.1 - 210.19) \\ &= 62,400 \text{ kJ/s} \Rightarrow \dot{m}_{\text{steam}} = 24.3 \text{ kg/s} \end{aligned}$$

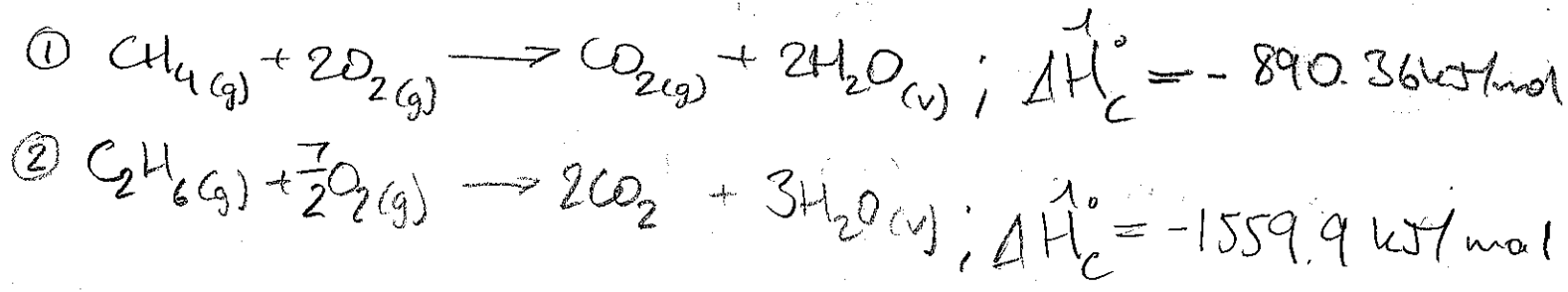
Our initial basis was 100 mol/s waste gas in,
so we can finally conclude that we generate
243 kg steam/mole waste gas

(5) a) Continuous Process ScenarioProcess Diagram:

Basis: 1 mol/s fuel gas



Reactions:



From our basis, we have .85 mol/s CH_4 & .15 mol/s C_2H_6 entering the furnace; we therefore need at least:

$$\frac{2 \text{ mol O}_2}{\text{mol CH}_4} \cdot \frac{.85 \text{ mol CH}_4}{\text{s}} + \frac{3.5 \text{ mol O}_2}{\text{mol C}_2\text{H}_6} \cdot \frac{.15 \text{ mol C}_2\text{H}_6}{\text{s}}$$

$$= 2.225 \text{ mol O}_2/\text{s} \dots$$

However, we know that the O_2 is in excess, so we can take a 10% excess, as a reasonable estimate...

(16)

$$10\% \text{ excess } \text{O}_2 = 1.1 (2.225) = 2.448 \text{ mol O}_2/\text{s}$$

Mass Balances:

$$\text{Carbon: } \dot{n}_2 = .85(1) + .15(2) \Rightarrow \dot{n}_2 = 1.15 \text{ mol CO}_2/\text{s}$$

$$\text{Hydrogen: } 2\dot{n}_3 = .85(4) + .15(6) \Rightarrow \dot{n}_3 = 2.15 \text{ mol H}_2\text{O}/\text{s}$$

$$\text{Oxygen: } \dot{n}_4 = .1(2.225) \Rightarrow \dot{n}_4 = .223 \text{ mol O}_2/\text{s}$$

$$\text{we also have } \dot{\xi}_1 = .85 \text{ \& } \dot{\xi}_2 = .15$$

Energy Balance:

Reference States: $\text{CH}_4(\text{g}), \text{C}_2\text{H}_6(\text{g}), \text{O}_2(\text{g}), \text{H}_2\text{O}(\text{l}), \text{ \& } \text{CO}_2(\text{g})$ (@ 25°C)

$$\Delta \hat{H}_{\text{f, H}_2\text{O}}^\circ = 44.01 \text{ kJ/mol}$$

Overall Balance:

Subst.	$\dot{n}_{\text{in}}$	$\hat{H}_{\text{in}}$	$\dot{n}_{\text{out}}$	$\hat{H}_{\text{out}}$
CH_4	.85	0	—	—
C_2H_6	.15	0	—	—
O_2	2.448	0	.223	0
CO_2	—	—	1.15	0
$\text{H}_2\text{O}(\text{v})$	—	—	2.15	44.01 (kJ/mol)

$$\dot{Q} = \dot{n}_{\text{CH}_4} (\Delta \hat{H}_c^\circ)_{\text{CH}_4} + \dot{n}_{\text{C}_2\text{H}_6} (\Delta \hat{H}_c^\circ)_{\text{C}_2\text{H}_6} + \sum_{\text{out}} \dot{n}_i \hat{H}_i - \sum_{\text{in}} \dot{n}_j \hat{H}_j$$

$$= .85(-890.36) \text{ kJ/s} + .15(-1559.9) \text{ kJ/s}$$

$$+ 2.15(44.01) \text{ kJ/s} = -896 \text{ kJ/s (kW)}$$

$$-\dot{Q} = 896 \text{ kW}$$

(heat from reactor)

note: heats of combustion are based on H_2O as liquid, whereas in the back of the

Murphy book H_2O as vapor is the reference point... If you

take $\text{H}_2\text{O}(\text{v})$ as your reference use $\Delta \hat{H}_f^\circ$ from Murphy? your answer should be

b) Constant Volume Scenario

Process Diagram, Basis, Reactions & Stoichiometry are the same as in part (a).

- ∴ We have 1 mol fuel gas $\Rightarrow .85 \text{ mol CH}_4$ & $.15 \text{ mol C}_2\text{H}_6$
- Theoretical $\text{O}_2 \Rightarrow 2.225 \text{ mol O}_2$
- 10% excess $\text{O}_2 \Rightarrow 2.448 \text{ mol O}_2$

Mass Balances:

Carbon: $n_2 = .85(1) + .15(2) = 1.15 \text{ mol CO}_2$

Hydrogen: $2n_3 = .85(4) + .15(6) \Rightarrow n_3 = 2.15 \text{ mol H}_2\text{O}$

Oxygen: $n_4 = .1(2.225) = .223 \text{ mol O}_2$

Energy Balance:

Reference States: $\text{CH}_4(\text{g}), \text{C}_2\text{H}_6, \text{O}_2(\text{g}), \text{H}_2\text{O}(\text{l}), \& \text{CO}_2(\text{g})$ (@ 25°C)

Since this is a constant-volume process, we use internal energies (which take into account the pressure-volume relation):

Subst.	n_{in}	U_{in}	n_{out}	U_{out}
CH_4	.85	0	—	—
C_2H_6	.15	0	—	—
O_2	2.448	0	.223	0
CO_2	—	—	1.15	0
$\text{H}_2\text{O}(\text{l})$	—	—	2.15	41.53 (kJ/mol)

$$\begin{aligned} \Delta \hat{U}_{\text{f}, \text{H}_2\text{O}}^\circ &= \Delta \hat{H}_{\text{f}, \text{H}_2\text{O}}^\circ - RT \\ &= 44.01 - \frac{8.314}{1000} \cdot 298\text{K} \\ &= 41.53 \text{ kJ/mol} \end{aligned}$$

(18)

$$\Delta \hat{U}_c^\circ = \Delta \hat{H}_c^\circ - RT \left(\sum_i \nu_{i,P} - \sum_j \nu_{j,R} \right)$$

$$(\Delta \hat{U}_c^\circ)_{\text{CH}_4} = (-890.36) \text{ kJ/mol} - 8.314 \times 10^{-3} \text{ kJ/mol} \cdot \text{K} \cdot 298 \text{ K} (1 + 2 - 1 - 2)$$

$$= -890.36 \text{ kJ/mol}$$

$$(\Delta \hat{U}_c^\circ)_{\text{C}_2\text{H}_6} = (-1559.9) - 8.314 \times 10^{-3} \cdot 298 (3 + 2 - 3.5 - 1)$$

$$= -1561.14 \text{ kJ/mol}$$

Overall Balance:

$$Q = \Delta U = n_{\text{CH}_4} (\Delta \hat{U}_c^\circ)_{\text{CH}_4} + n_{\text{C}_2\text{H}_6} (\Delta \hat{U}_c^\circ)_{\text{C}_2\text{H}_6} + \sum_{\text{out}} n_i \hat{U}_i - \sum_{\text{in}} n_j \hat{U}_j$$

$$= .85(-890.36) + .15(-1561.14) + 2.15(41.53)$$

$$= -902 \text{ kJ}$$

or

$$-Q = 902 \text{ kJ transferred from reactor}$$

c) Because the O_2 (& N_2 , if air is used) are at the same temperature & pressure at both the inlet & the outlet, their enthalpies & internal energies do not change, and also zero, since we took these conditions as reference, the amounts do not make a difference in the overall energy balance as long as there is enough O_2 for complete combustion of the fuel gas...